

Qatar-1b

R-0305

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_171008 · created 2026-07-17T21:34:41+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458034.6938 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.014
Transit depth (Rp/Rs)^2	2.08 +/- 0.41 [%]
Orbital Inclination (inc)	87.46 +/- 1.22
Airmass coefficient 1 (a1)	0.968 +/- 0.014
Airmass coefficient 2 (a2)	0.023 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#1 - [206, 232]
Optimal Aperture	3.49
Optimal Annulus	5.22
Transit Duration (day)	0.0801 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.014 vs 0.1463 (deviation 0.16σ)
Duration fit vs published	0.0801 d vs 0.0692 d (deviation 3.77σ)
Mid-time O–C	-0.5 min
Depth SNR	10.3
Residual scatter	1.5 %

Light curve

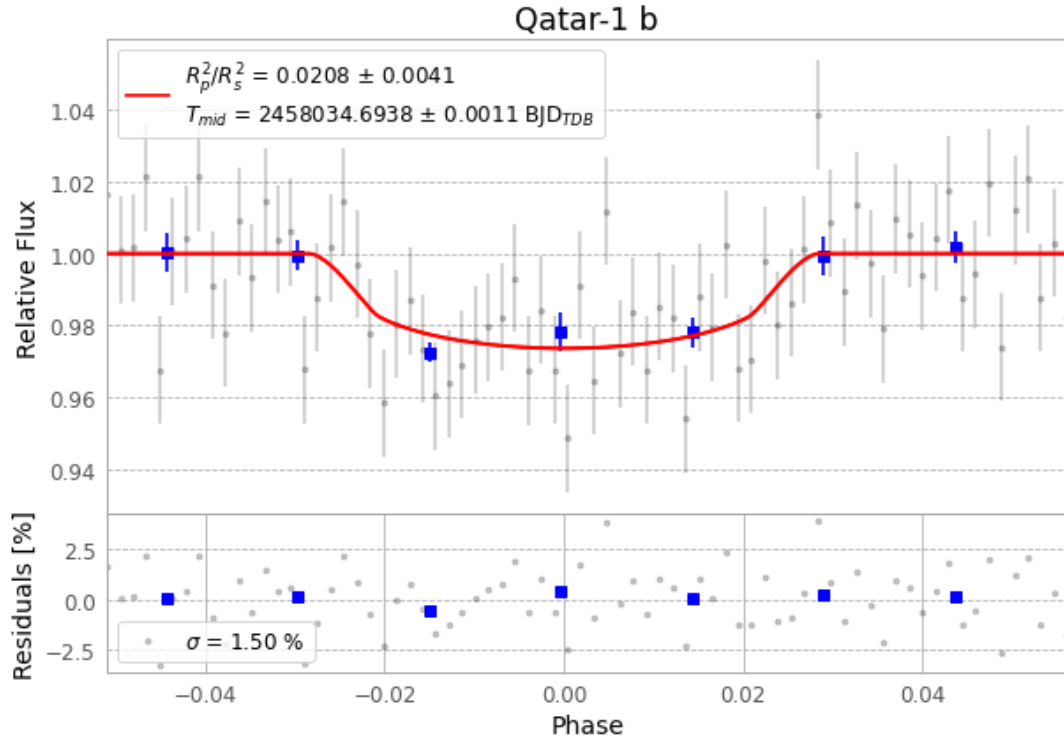


Figure 1 — FinalLightCurve_Qatar-1 b_2017-10-07.png (SHA-256 0bbd2c22c95959cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 305], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50b417e448850180...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

74 FITS frames (49 MB), Qatar-1171008025112.FITS → Qatar-1171008063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-171008.log (74afede75e7b...), FinalLightCurve_Qatar-1 b_2017-10-07.png (0bbd2c22c959...), FinalLightCurve_Qatar-1 b_2017-10-07.csv (c04713442779...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 21:34Z.